

MMI 806 · PRINCIPLES OF RESPONSIBLE AI

MSc in Mathematical Innovation · Open University of Kenya

Module 9: Looking to the Future

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LEARNING OUTCOMES

1. Describe and distinguish the four categories of future AI developments: Nevers, Whens, Ifs, and Unknowns.
2. Evaluate which AI applications are technically realistic, highly speculative, or ethically impermissible.
3. Explore how assumptions about future AI shape current decisions in education, policy, and society.
4. Anticipate potential breakthroughs and ethical dilemmas in the medium and long-term future of AI.

1 Module Overview

Responsible AI is not only about the present — it is about preparing for futures that are not yet certain. Decisions made today (about data infrastructure, regulatory frameworks, educational curricula) will shape whether future AI systems are beneficial or harmful. This module develops a disciplined approach to AI futures thinking, distinguishing what is certain from what is speculative and what is impermissible from what is merely unknown.

THE DANGER OF AI FUTURES HYPE

AI is subject to extraordinary hype cycles — periods of extreme optimism followed by disillusionment. The first AI summer (1960s), second AI summer (1980s), and current deep learning revolution all followed similar patterns. Responsible AI thinking requires sceptical engagement with both utopian promises and apocalyptic fears, grounded in what is actually technically understood.

At the same time, dismissing all speculative AI developments as science fiction is equally irresponsible. The harms of present AI systems were foreseeable and forecast by researchers who were systematically ignored. Responsible preparation demands taking plausible futures seriously.

2 The Four-Category Framework

FOUR CATEGORIES OF FUTURE AI DEVELOPMENTS

Nevers: AI applications that are technically or physically impossible, or ethically prohibited regardless of technical feasibility. These will not and should not happen.

Whens: AI applications that are technically feasible and will almost certainly occur; the question is when, not whether. These require active preparation.

Ifs: AI applications that are technically plausible but uncertain; outcomes depend on scientific breakthroughs, regulatory decisions, or social choices. These require careful monitoring and contingency planning.

Unknowns: Developments we cannot currently conceptualise or predict. These require epistemic humility and adaptive governance frameworks that can respond to surprises.

3 Nevers: What AI Will Not and Should Not Do

EXAMPLES: The Nevers

Truly Never (technically impossible or physically meaningless):

- A perfect, omniscient predictive system (violates computational limits and inherent unpredictability of complex systems).
- A system that genuinely “understands” in the way humans understand — consciousness and qualia remain philosophically and scientifically unresolved.

Should Never (ethically prohibited regardless of feasibility):

- AI systems designed to target autonomous lethal force at individuals without human authorisation (autonomous weapons “killer robots”).
- AI-powered social scoring systems that rank citizens and restrict their freedoms based on predicted behaviour (as piloted in China’s Social Credit System in some cities).
- AI systems designed to manipulate electoral choices through personalised psychological targeting at scale.
- AI systems providing medical diagnoses with no pathway for professional oversight in life-threatening conditions.

The EU AI Act and the “Nevers”: The Act’s “unacceptable risk” category attempts to codify the ethical Nevers into law. Its prohibitions include real-time biometric surveillance, social scoring, and subliminal manipulation.

4 Whens: Developments Requiring Active Preparation

EXAMPLES: The Whens

- **AI-powered autonomous vehicles:** Already partially deployed; full deployment at scale is a matter of regulatory approval and infrastructure, not technical feasibility. *Preparation:* liability law reform; infrastructure investment; worker displacement planning.
- **AI-driven drug discovery:** Already producing novel candidate compounds; clinical validation pipeline will determine deployment timescale. *Preparation:* regulatory frameworks for AI-discovered drugs; data sharing agreements.
- **AI in judicial decision-making:** Already used in sentencing (COMPAS); will expand. *Preparation:* transparency requirements; contestability rights; training for judiciary.
- **AI replacing large categories of cognitive labour:** Routine white-collar tasks (document processing, basic coding, transcription) are already substantially automated. *Preparation:* educational curriculum reform; social safety nets; retraining programmes.
- **AI-powered personalised healthcare in Africa:** Smartphone penetration, mobile health infrastructure, and disease burden make AI-powered diagnostics a high-value near-term deployment in East Africa. *Preparation:* regulatory frameworks; training data collection; practitioner integration.

5 Ifs: Uncertain but Consequential Possibilities

EXAMPLES: The Ifs

- **Artificial General Intelligence (AGI):** A system with human-level cognitive capabilities across all domains. No current system is close to this; it may require fundamental scientific breakthroughs we cannot currently see. Some researchers believe it will never occur; others believe it is decades away. *If it occurs:* the governance implications are profound and require active current preparation.
- **AI consciousness and rights:** If AI systems develop genuine sentience (a scientifically and philosophically unresolved question), what moral status should they have? This is currently speculative but the question is not trivially dismissible.
- **AI-driven scientific revolution:** AlphaFold demonstrated that AI can solve decade-old scientific problems in biology. Whether AI will unlock comparable breakthroughs in physics, mathematics, and climate science is uncertain.
- **AI in governance:** AI advisory systems for policy decisions are already deployed. Whether AI will be given *decision-making* authority in governance is a social and legal choice, not a technical one.

6 Unknowns: Epistemic Humility and Adaptive Governance

WHAT WE DON'T KNOW WE DON'T KNOW

The most important AI developments may be ones we cannot currently conceptualise. In 2000, few predicted: (1) that social media algorithms would become the primary mechanism for political radicalisation at scale; (2) that smartphone proliferation would give AI access to the intimate data of billions of people in low-income countries within 15 years; (3) that generative AI would make high-quality synthetic media indistinguishable from real media.

Adaptive governance — frameworks that can respond to technological surprises rather than only governing anticipated applications — is therefore essential. Principle-based regulation (governing outcomes and values) is more resilient to technological unknowns than rule-based regulation (governing specific applications).

7 Implications for Education and Society in Africa

LOOKING FORWARD: AI and Kenyan Education

The “Whens” for Kenyan education include:

- AI tutoring systems adapting to individual student needs: opportunity for personalised learning at scale in large classes; risk of reduced human interaction and teacher deskilling.
- AI-powered examination integrity systems: facial recognition and behavioural monitoring during remote exams; significant privacy and equity implications (high-quality cameras required; works less well for darker skin tones).
- AI-generated curriculum content: efficiency gains; risk of content misaligned with Kenyan cultural context; risk of over-dependence on non-Kenyan AI systems.
- Automated grading of complex responses: frees teacher time; introduces risk of systematic grading bias; may disadvantage students who write in non-standard English.

Key question: How should Kenya’s education sector prepare for these “Whens” to maximise benefits and minimise harms?

8 Module Assignment: Forecasting AI Reflection Portfolio

MODULE 9 ASSIGNMENT

1. **Choose one category:** Nevers, Whens, Ifs, or Unknowns.
2. **Choose one specific AI development** within that category.
3. **Contextual impact reflection:** How would this development affect Kenya or East Africa specifically? Who would benefit? Who would be harmed? What structural conditions in Kenya affect the likelihood or nature of this impact?
4. **Preparedness reflection:** What decisions should be made now — by government, educational institutions, civil society, or individuals — to ensure this development is handled responsibly? Write 300–400 words.

9 Key Concepts and Terminology

Nevers	AI developments that are technically impossible or ethically prohibited.
Whens	AI developments that will almost certainly occur; timing uncertain, not occurrence.
Ifs	AI developments that are plausible but dependent on unresolved scientific or social factors.
Unknowns	AI developments currently unconceptualised; require adaptive governance.
AGI	Artificial General Intelligence: hypothetical system with human-level cognitive capability across all domains.
Adaptive Governance	Regulatory frameworks designed to respond to technological surprises, not only anticipated applications.
Hype Cycle	Pattern of inflated expectations, disillusionment, and eventual realistic assessment in technology adoption.

10 Revision Questions

1. Classify each of the following using the four-category framework and justify your classification: (a) AI that perfectly predicts individual human behaviour 10 years in advance; (b) AI routinely performing routine legal document review; (c) AI achieving consciousness; (d) AI replacing all primary school teachers globally.

2. The EU AI Act prohibits “subliminal manipulation” by AI. How would you define this? Give an example of an AI system that might violate this prohibition and one that might not.
3. AGI (Artificial General Intelligence) is classified as an “If” in this module. Some leading AI researchers believe it will be achieved within 10 years; others believe it will never occur. What is the responsible institutional response to this uncertainty?
4. Describe the concept of an AI “hype cycle.” Identify one specific AI technology that has gone through a hype cycle and describe the phases it went through.
5. What does “adaptive governance” mean for responsible AI in Kenya? What specific institutional arrangements would enable adaptive governance rather than rule-based regulation?
6. The development of autonomous lethal weapons is classified as a “Should Never.” Construct the strongest argument *for* autonomous weapons, then rebut it from a responsible AI perspective.
7. Identify one “Unknown” in AI development that you believe is plausible but currently underappreciated. Describe what it is, why it is currently invisible, and what its implications for responsible AI governance might be.

MODULE 9 SUMMARY

- The four-category framework — Nevers, Whens, Ifs, Unknowns — provides a disciplined tool for AI futures thinking, distinguishing certainty from speculation and prohibition from possibility.
- Responsible preparation requires active planning for “Whens” (certain developments), careful monitoring of “Ifs” (uncertain ones), and adaptive governance for “Unknowns.”
- AI development is subject to hype cycles; responsible thinking requires scepticism about both utopian and dystopian extremes.
- The “Nevers” — both technically impossible and ethically prohibited — must be clearly articulated and enforced through regulation.
- Kenyan and African contexts will experience AI futures differently from high-income country contexts: different infrastructure, regulatory capacity, data availability, and social conditions.
- *The future of AI is not predetermined. It is the product of decisions being made now — by researchers, policymakers, companies, and communities. Responsible AI education prepares you to participate meaningfully in those decisions.*

11 Further Reading

- **Russell, S.** (2019). *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking.
- **Bostrom, N.** (2014). *Superintelligence*. Oxford University Press. — Read critically.
- **Floridi, L. et al.** (2020). An ethical framework for a good AI society. *Minds and Machines*, 28, 689–707.

- **UNESCO.** (2021). Recommendation on the Ethics of Artificial Intelligence. <https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>